Task 3 — Learn .gitignore

DEVOPS-005: Git Advanced Workflow & Team Collaboration

Step 1: Create files

```
.env
password.txt
database.log
debug.log
temp.txt
```

Step 2: Edit .gitignore — add

```
.env
*.log
password.txt
temp/
```

Step 3: Run

```
git status
```

Result: .env, password.txt, database.log, and debug.log no longer appear as untracked files, because they are matched by the .gitignore rules. Note: temp.txt still appears as untracked, since the rule temp/ only matches a folder named temp, not a file named temp.txt.

Step 4: Stage and verify

```
git add .
git status
```

Result: All non-ignored files are staged. The ignored files remain absent from the staged list, confirming .gitignore is working correctly.

Important Question

Why should passwords and API keys never be committed to Git?

Git keeps a permanent history of every commit. Once a secret is committed, it remains in the repository's history even if it is deleted in a later commit, unless the history itself is rewritten (using tools like git filter-repo or BFG), which is risky and disruptive, especially on a shared repository.

If the repository is ever pushed to a public or shared remote such as GitHub, the exposed secret becomes visible to anyone with access, including forks, clones, and cached copies of the repo.

Leaked credentials can allow attackers to access databases, cloud services, or third-party APIs, potentially leading to data breaches, service abuse, or financial loss.

For this reason, secrets should be kept in environment variables, .env files excluded via .gitignore, or a dedicated secrets manager — never hardcoded or committed directly into source control.